

Lab Experiment: 7

Title: MongoDB install and Run command

Course title: Data Mining Problem Description

Course code: CSE-452
4th Year 1st Semester Examination 2025

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Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

cluster14584.5wca2wu.mongodb.net

- admin
- config
- local
- sample_airbnb
- sample_analytics
- sample_geospatial
- sample_guides
- sample_mflix
- sample_restaurants
- sample_supplies
- sample_training
- sample_weatherdata
- studentDB

Welcome mongosh: cluster14584...

> MONGOSH

```
> db.students.insertMany([
  { name: "Ali", age: 20, deptId: 1 },
  { name: "Sara", age: 22, deptId: 2 },
  { name: "Rafi", age: 21, deptId: 1 },
  { name: "Sakib", age: 23, deptId: 1 },
  { name: "Nusrat", age: 22, deptId: 2 },
  { name: "Tanvir", age: 24, deptId: 1 },
  { name: "Mim", age: 21, deptId: 2 },
  { name: "Rasel", age: 25, deptId: 1 },
  { name: "Tania", age: 20, deptId: 2 },
  { name: "Fahim", age: 23, deptId: 1 },

  { name: "Jony", age: 22, deptId: 2 },
  { name: "Lamia", age: 21, deptId: 1 },
  { name: "Rakib", age: 24, deptId: 2 },
  { name: "Nayeen", age: 23, deptId: 1 },
  { name: "Sumi", age: 20, deptId: 2 },
  { name: "Hasib", age: 25, deptId: 1 },
  { name: "Opi", age: 22, deptId: 2 },
  { name: "Imran", age: 24, deptId: 1 },
  { name: "Shanto", age: 23, deptId: 2 },
  { name: "Mahin", age: 21, deptId: 1 }

])

< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d74929178cc73a14a5b6e0'),
    '1': ObjectId('69d74929178cc73a14a5b6e1'),
    '2': ObjectId('69d74929178cc73a14a5b6e2'),
    '3': ObjectId('69d74929178cc73a14a5b6e3'),
    '4': ObjectId('69d74929178cc73a14a5b6e4'),
    '5': ObjectId('69d74929178cc73a14a5b6e5'),
    '6': ObjectId('69d74929178cc73a14a5b6e6'),
    '7': ObjectId('69d74929178cc73a14a5b6e7'),
    '8': ObjectId('69d74929178cc73a14a5b6e8'),
    '9': ObjectId('69d74929178cc73a14a5b6e9'),
    '10': ObjectId('69d74929178cc73a14a5b6ea'),
    '11': ObjectId('69d74929178cc73a14a5b6eb'),
    '12': ObjectId('69d74929178cc73a14a5b6ec'),
    '13': ObjectId('69d74929178cc73a14a5b6ed'),
    '14': ObjectId('69d74929178cc73a14a5b6ee'),
    '15': ObjectId('69d74929178cc73a14a5b6ef'),
    '16': ObjectId('69d74929178cc73a14a5b6f0'),
    '17': ObjectId('69d74929178cc73a14a5b6f1'),
    '18': ObjectId('69d74929178cc73a14a5b6f2'),
    '19': ObjectId('69d74929178cc73a14a5b6f3')
  }
}
```

Atlas atlas-dbivw3-shard-0 [primary] studentDB>

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{ } My Queries

🏗 Data Modeling

CONNECTIONS (1)

✕ + ...

Search connections

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- ▶ 📁 sample_mflix
- ▶ 📁 sample_restaurants
- ▶ 📁 sample_supplies
- ▶ 📁 sample_training
- ▶ 📁 sample_weatherdata
- ▼ 📁 studentDB

Welcome

➤ mongosh: cluster14584....

+

> _MONGOSH

```
<
> db.students.countDocuments()
< 20
> db.students.find()
< {
  _id: ObjectId('69d74929178cc73a14a5b6e0'),
  name: 'Ali',
  age: 20,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 22,
  deptId: 2
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e2'),
  name: 'Rafi',
  age: 21,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e3'),
  name: 'Sakib',
  age: 23,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e4'),
  name: 'Nusrat',
  age: 22,
  deptId: 2
}
}
```

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🔗 Data Modeling

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▶ config

▶ local

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▶ sample_guides

▶ sample_mflix

▶ sample_restaurants

▶ sample_supplies

▶ sample_training

▶ sample_weatherdata

▼ studentDB

Welcome

> mongosh: cluster14584...



>_MONGOSH

> db.students.find().pretty()

```
< {
  _id: ObjectId('69d74929178cc73a14a5b6e0'),
  name: 'Ali',
  age: 20,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 22,
  deptId: 2
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e2'),
  name: 'Rafi',
  age: 21,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e3'),
  name: 'Sakib',
  age: 23,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e4'),
  name: 'Nusrat',
  age: 22,
  deptId: 2
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e5'),
```

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🔧 Data Modeling

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- sample_supplies
- sample_training
- sample_weatherdata
- studentDB

Welcome mongosh: cluster14584....

> _MONGOSH

```
}  
  
{  
  _id: ObjectId('69d74929178cc73a14a5b6f3'),  
  name: 'Mahin',  
  age: 21,  
  deptId: 1  
}  
  
> db.students.find({age:23})  
< {  
  _id: ObjectId('69d74929178cc73a14a5b6e3'),  
  name: 'Sakib',  
  age: 23,  
  deptId: 1  
}  
  
{  
  _id: ObjectId('69d74929178cc73a14a5b6e9'),  
  name: 'Fahim',  
  age: 23,  
  deptId: 1  
}  
  
{  
  _id: ObjectId('69d74929178cc73a14a5b6ed'),  
  name: 'Nayeem',  
  age: 23,  
  deptId: 1  
}  
  
{  
  _id: ObjectId('69d74929178cc73a14a5b6f2'),  
  name: 'Shanto',  
  age: 23,  
  deptId: 2  
}
```

Atlas atlas-dbivv3-shard-0 [primary] studentDB>

```
> db.students.insertOne({name:"Khaled",age:26,deptId:391})
< {
  acknowledged: true,
  insertedId: ObjectId('69d74afc178cc73a14a5b6f4')
}
> db.students.find({deptId:391})
< {
  _id: ObjectId('69d74afc178cc73a14a5b6f4'),
  name: 'Khaled',
  age: 26,
  deptId: 391
}
Atlas atlas-dbivv3-shard-0 [primary] studentDB > |
```

```
> db.students.updateOne(
  { name: "Khaled" },
  { $set: { age: 30 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.find({deptId:391})
< {
  _id: ObjectId('69d74afc178cc73a14a5b6f4'),
  name: 'Khaled',
  age: 30,
  deptId: 391
}
Atlas atlas-dbivv3-shard-0 [primary] studentDB > |
```

```
> db.students.updateOne(
  { name: "Sara" },
  { $set: { deptId: 3 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.students.find({name:"Sara"})
< {
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 22,
  deptId: 3
}
Atlas atlas-dbivv3-shard-0 [primary] studentDB> |
```

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Welcome

mongosh: cluster14584....



>_MONGOSH

```
> db.students.updateMany(
  { age: 22 },
  { $set: { status: "active" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
> db.students.find()
< {
  _id: ObjectId('69d74929178cc73a14a5b6e0'),
  name: 'Ali',
  age: 20,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 22,
  deptId: 3,
  status: 'active'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e2'),
  name: 'Rafi',
  age: 21,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e3'),
  name: 'Sahil',
  age: 22,
  deptId: 1
}
```

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{ } My Queries

🔗 Data Modeling

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- ▶ admin
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- ▶ sample_supplies
- ▶ sample_training
- ▶ sample_weatherdata
- ▼ studentDB

Welcome

➤ mongosh: cluster14584....



> _MONGOSH



```
> db.students.updateMany(
  {},
  { $inc: { age: 1 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 21,
  modifiedCount: 21,
  upsertedCount: 0
}
> db.students.find()
< {
  _id: ObjectId('69d74929178cc73a14a5b6e0'),
  name: 'Ali',
  age: 21,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 23,
  deptId: 3,
  status: 'active'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e2'),
  name: 'Rafi',
  age: 22,
  deptId: 1
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e3'),
  name: 'Sahib'
}
```

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{ } My Queries

🔗 Data Modeling

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- sample_supplies
- sample_training
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Welcome mongosh: cluster14584....

```
> _MONGOSH
> db.students.updateMany(
  { deptId: 1 },
  { $set: { group: "CSE" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 11,
  modifiedCount: 11,
  upsertedCount: 0
}
> db.students.find()
< {
  _id: ObjectId('69d74929178cc73a14a5b6e0'),
  name: 'Ali',
  age: 21,
  deptId: 1,
  group: 'CSE'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 23,
  deptId: 3,
  status: 'active'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e2'),
  name: 'Rafi',
  age: 22,
  deptId: 1,
  group: 'CSE'
}
{
```

```
> db.students.find({name:"Rafi"})
< {
  _id: ObjectId('69d74929178cc73a14a5b6e2'),
  name: 'Rafi',
  age: 22,
  deptId: 1,
  group: 'CSE'
}
> db.students.deleteOne({name:"Rafi"})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.students.find({name:"Rafi"})
<
Atlas atlas-dbivv3-shard-0 [primary] studentDB > |
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

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- sample_guides
- sample_mflix
- sample_restaurants
- sample_supplies
- sample_training
- sample_weatherdata
- studentDB

Welcome mongosh: cluster14584....

```
>_MONGOSH
}
> db.students.find().sort({ age: -1 })
< {
  _id: ObjectId('69d74afc178cc73a14a5b6f4'),
  name: 'Khaled',
  age: 31,
  deptId: 391
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e7'),
  name: 'Rasel',
  age: 26,
  deptId: 1,
  group: 'CSE'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6ef'),
  name: 'Hasib',
  age: 26,
  deptId: 1,
  group: 'CSE'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6e5'),
  name: 'Tanvir',
  age: 25,
  deptId: 1,
  group: 'CSE'
}
{
  _id: ObjectId('69d74929178cc73a14a5b6f1'),
  name: 'Imran',
  age: 25,
  deptId: 1,
```

Compass

{ } My Queries

🔗 Data Modeling

CONNECTIONS (1)

✕ + ...

Search connections



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- ▶ 🗄 sample_mflix
- ▶ 🗄 sample_restaurants
- ▶ 🗄 sample_supplies
- ▶ 🗄 sample_training
- ▶ 🗄 sample_weatherdata
- ▼ 🗄 studentDB

Welcome ➤ mongosh: cluster14584.... +

> _MONGOSH



> db.students.find().limit(5)

< {

```
  _id: ObjectId('69d74929178cc73a14a5b6e0'),
  name: 'Ali',
  age: 21,
  deptId: 1,
  group: 'CSE'
}
```

{

```
  _id: ObjectId('69d74929178cc73a14a5b6e1'),
  name: 'Sara',
  age: 23,
  deptId: 3,
  status: 'active'
}
```

{

```
  _id: ObjectId('69d74929178cc73a14a5b6e3'),
  name: 'Sakib',
  age: 24,
  deptId: 1,
  group: 'CSE'
}
```

{

```
  _id: ObjectId('69d74929178cc73a14a5b6e4'),
  name: 'Nusrat',
  age: 23,
  deptId: 2,
  status: 'active'
}
```

{

```
  _id: ObjectId('69d74929178cc73a14a5b6e5'),
  name: 'Tanvir',
  age: 25,
```